

ANALYSIS2 - LIST 2 : FUNCTION SERIES

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Function series and their convergence

1. Definition of a pointwise convergence of a sequence of functions.
2. Definition of a uniform convergence of a sequence of functions.
3. Example of a function sequence of continuous function which is pointwise convergent to a discontinuous function.
4. Cauchy's condition of uniform convergence.
5. Definition of a function series.
6. Theorem about the differentiability of a function series.
7. Example of the pointwise convergent function series to which the theorem about the differentiability of a function series does not apply.
8. Theorem about integration of a function series.
9. Example of the pointwise convergent function series to which theorem about the integration of a function series does not apply.
10. Weierstrass test of uniform convergence of a function series.

Power series

1. Definition of the radius of convergence of a power series.
2. Theorem about the convergence of a power series.
3. Definition of the absolute convergence.
4. Where a power series is absolutely and uniformly convergent?
5. Cauchy's convergence test.
6. D'Alembert's convergence test.
7. Check the convergence of a given power series.